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**INVERSE HEAT TRANSFER ANALYSIS OF
SUBCOOLED FLOW BOILING IN MINICHANNELS
USING TREFFTZ FUNCTIONS AND TIKHONOV
REGULARIZATION**

BEATA MACIEJEWSKA, MAGDALENA PIASECKA, BARTŁOMIEJ SKORUPSKI
Politechnika Świętokrzyska
Al. Tysiąclecia Państwa Polskiego 7, 25-314 Kielce
e-mail: beatam@tu.kielce.pl

The main objective of this study was to perform heat transfer calculations for flow boiling in minichannels based on experimental data, using Trefftz functions and Tikhonov regularization. In the regularization process, Morozov's discrepancy principle was applied, taking into account the uncertainty of temperature measurements estimated using the Monte Carlo method. The experimental setup consisted of five parallel minichannels with a depth of 1 mm, sharing a common heated wall. Local heat transfer coefficients at the interface between the working fluid and the heated wall were determined using the finite element method with Trefftz-type basis functions. The regularization parameter was determined according to Morozov's discrepancy principle, assuming the noise level to be equal to the estimated uncertainty of temperature measurements. In particular, it was selected so that the norm of the residual was comparable to the noise level in the experimental data, which prevents overfitting to measurement disturbances and ensures consistency with the quality of the available data.